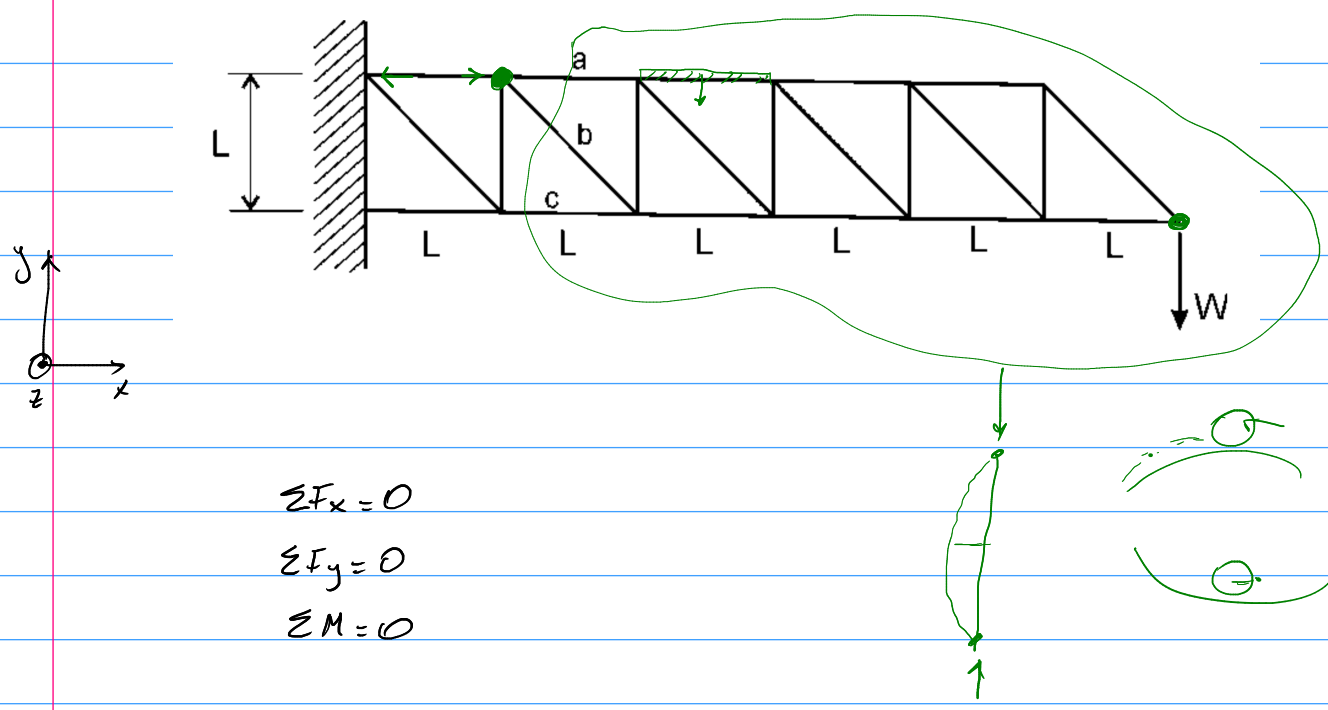
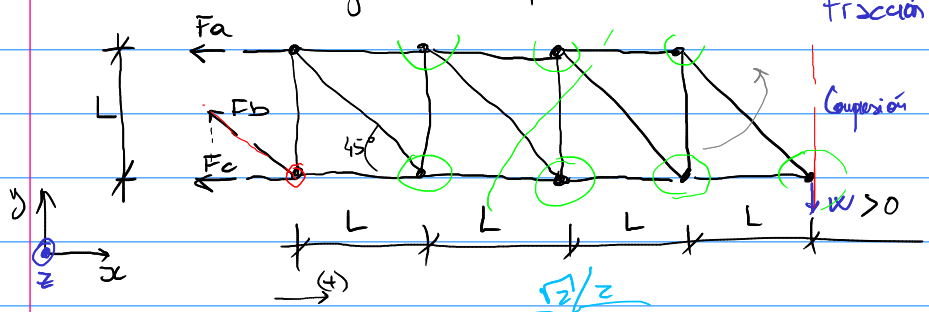


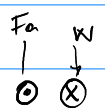
$\Sigma F_x = 0$

$\Sigma F_y = 0$

$\Sigma M = 0$

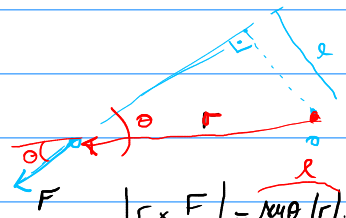
Diagrama de cuerpo libre





$$\Sigma F_x = -F_a - F_b \cdot \underbrace{\cos(45°)}_{\sqrt{2}/2} - F_c = 0$$

$$\Sigma F_y = F_b \cdot \underbrace{\operatorname{sen}(45°)}_{\sqrt{2}/2} - W = 0 \Rightarrow F_b = \frac{2W}{\sqrt{2}}$$



$|\underline{r} \times \underline{F}| = \overbrace{\operatorname{sen}\theta\,|r|}^{\ell} \cdot |F|$

brazo de palanca

$$\Sigma M = F_a L - W\,4 \cdot L = 0 \Rightarrow F_a = 4W$$

$$F_c = -F_a - F_b \cdot \frac{\sqrt{2}}{2} = -5W$$

3D)

$\Sigma F_x = 0$

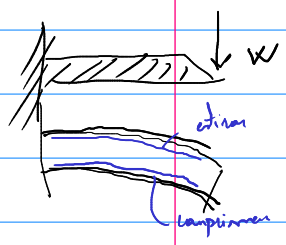
$\Sigma F_y = 0$

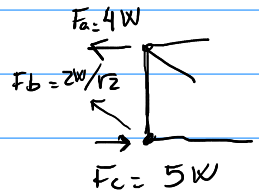
$\Sigma F_z = 0$

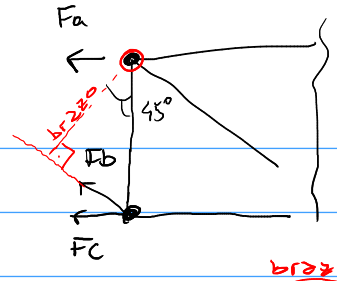
$\Sigma M_x = 0$

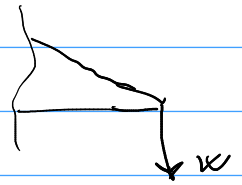
$\Sigma M_y = 0$

$\Sigma M_z = 0$





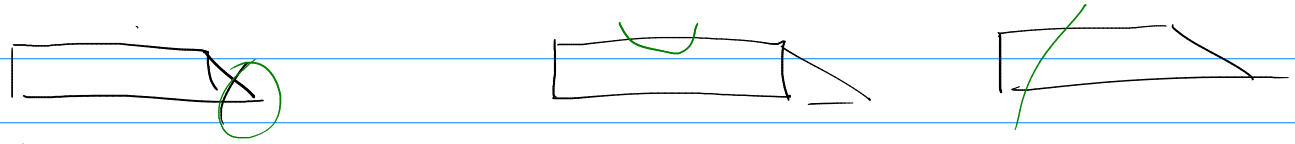


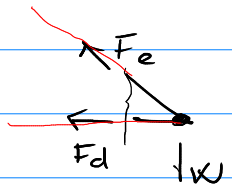


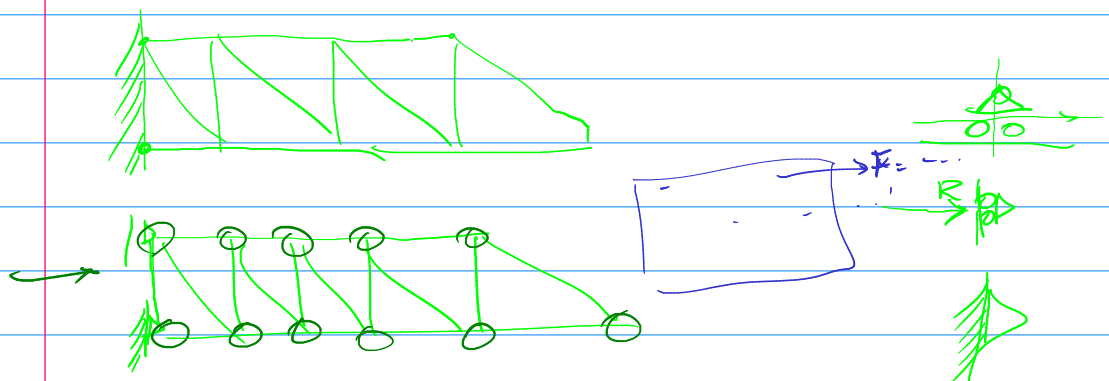
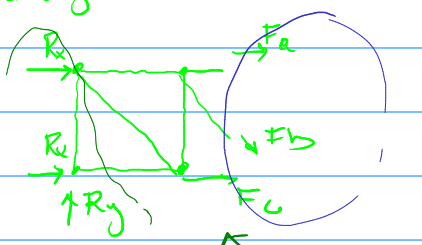
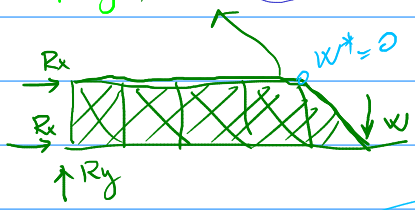
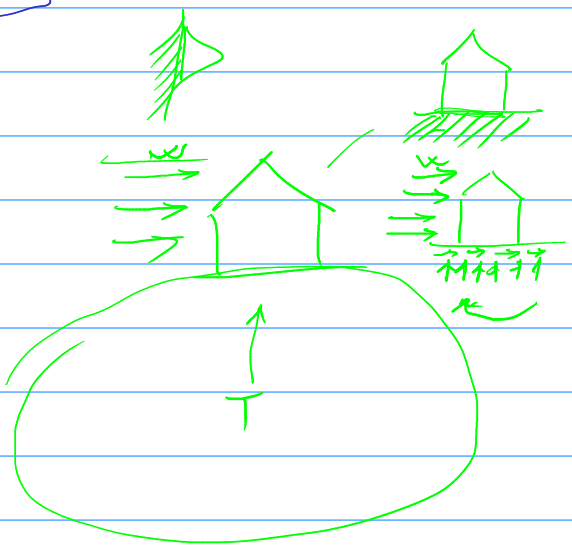
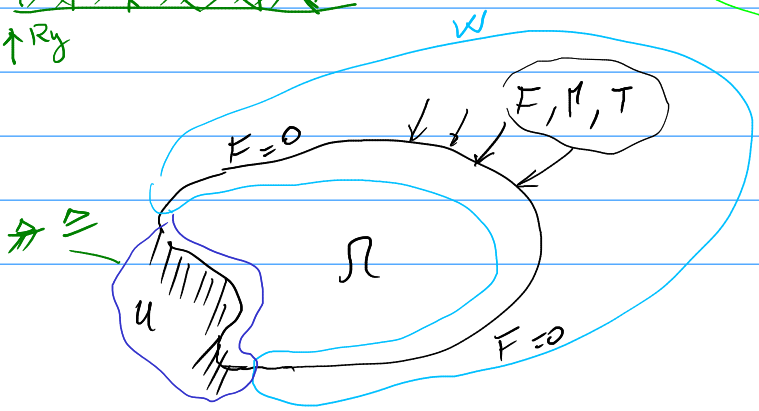
$$\sum M' = -F_c \cdot L - F_b \cdot \overbrace{\cos 45^\circ}^{\text{braço}} \cdot L - W\,4L$$

$$= -(-5W) \cdot L - \frac{2W}{\sqrt{2}} \cdot \frac{\sqrt{2}}{2} \cdot L - 4WL$$

$$= 5WL - WL - 4WL = 0 \Rightarrow \text{Verifica}$$







Diagrama